

Homework (Jalapeno)

Q1. Evaluate the expression $2 \cdot 3 + 12 \div 4$.

- (A) 9
- (B) 10
- (C) 11
- (D) 12
- (E) 13

Q2. Which of the following expressions represents the phrase '5 more than 3 times a number'? Let x be the number.

- (A) $3x + 5$
- (B) $3x - 5$
- (C) $5x + 3$
- (D) $5x - 3$
- (E) $x + 3$

Q3. Simplify the expression $2x + 5x$.

- (A) $7x$
- (B) $7x + 1$
- (C) $7x - 1$
- (D) $x + 1$
- (E) $x - 1$

Q4. In ancient Egypt, a rectangle's area is $A = 2x^2 + 5x$. If $x = 3$, what is the area?

- (A) 27
- (B) 31
- (C) 36
- (D) 41
- (E) 46

Q5. What is the value of x in the equation $x + 2 = 7$?

- (A) 3
- (B) 5
- (C) 7
- (D) 9
- (E) 11

Q6. The ancient Babylonians used the expression $3 \cdot x - 2$ to calculate the area of a triangle. If the area is 7, what is x ?

- (A) 3
- (B) 5
- (C) 7
- (D) 9
- (E) 11

Q7. The ancient Greeks used the expression $2x^2 + x$ to model the growth of a population. If $x = 2$, what is the population?

- (A) 10
- (B) 12
- (C) 14
- (D) 16
- (E) 18

Q8. Evaluate the expression $(2 + 3) \cdot (4 - 1)$.

- (A) 10
- (B) 12
- (C) 14
- (D) 15
- (E) 18

Open Ended Questions (FRQ)**Q9.** The ancient Egyptians built a rectangular temple with an area of $A = 2x^2 + 5x$. If the length of the temple is $x + 2$, find the width of the temple in terms of x . Provide your answer in the space below.**Q10.** The ancient Greeks used the expression $3 \cdot x - 2$ to calculate the area of a triangle. If the area is 7, find the value of x and use it to calculate the perimeter of the triangle if the base is $x + 1$ and the height is $x - 1$. Provide your answer in the space below.**Chef's Tips**

- Tip 1: Ensure students understand the order of operations (PEMDAS).
- Tip 2: Encourage students to simplify expressions before evaluating them.
- Tip 3: Remind students to check their units and ensure they are consistent throughout the problem.